Can AI Be Creative?

Podcast

Summary

This podcast episode explores the question "Is AI capable of creativity"? Hosts Alona Fyshe and Scott Lilwall interview Matthew Guzdial, an Amii fellow, and Canada CIFAR AI Chair at the University of Alberta. Guzdial researches computational creativity which is described as the study of how human notions of creativity can be modelled and supported by computers. His goal is to use his background in art and animation combined with computer science to make creative technical tools accessible to everyone.Guzdial points out that technology has been made accessible in specific creative areas, for instance, in creative writing but that this same accessibility still ceases to exist in more complex areas like video game development. He believes that anyone that has an idea, should be able to access the tools required to create it.

Guzdial dives into his work on mixed-initiative collaboration, where both humans and AI play active roles in creating something new. He discusses this in his project, "More AI Maker", a level design tool where a human and AI partner take turns adding elements to a game world. The system learns to adapt to each user's unique style, whether that style is more logical and structured or playful and unexpected, demonstrating how AI can shift from being a passive tool to an active co-creator in creative endeavors.

Guzdial refers to cognitive scientist Margaret Boden's definition of creativity and how this definition includes three types: exploratory, combinational and transformational. He also introduces the distinction between P-creativity (personal creativity) and H-creativity (historical creativity) and the discussion concludes that AI generated content is often "p-creative" for the user, even if it's not "h-creative" (novel throughout history).

Finally the podcast addresses potential issues when AI is merged into more creative industries. Issues with data consent and job displacement are acknowledged but Guzdial emphasizes the importance of a collaborative approach and using AI as a tool to support humans in the creative process, rather than replace them. Although game development is a major focus in the episode, the ideas discussed apply to any creative process. Overall, the discussion highlights that AI has the potential to act as a collaborator across disciplines, expanding on what's possible and supporting human creativity in transformative ways.

Key Terms

Term	Definition
Computational Creativity	The practice of using computers to simulate, model, or enhance creative processes.
Mixed-Initiative	The idea that both a human and a computational system can take initiative in a collaborative creative process.
P-Creativity (Personal Creativity)	When an individual judges a work to be creative because they have never seen anything like it before.
H-Creativity (Historical Creativity)	A theoretical concept that refers to a work being considered creative across all of history.
Transfer Learning	The process of taking knowledge from one domain with a lot of data and applying it to another domain with less data.
Explainable AI (XAI)	The part of AI research that focuses on making the decision-making process of AI systems understandable to humans.
Deep Reinforcement Learning (Deep RL)	This is a type of machine learning where an AI agent learns through a system of rewards and penalties.
Human-Centered Design	This is a design approach that puts the needs and perspectives of the user at the forefront of the development process.

3 Key Takeaways

1. AI can be your creative partner, guiding you from idea to execution without requiring deep technical expertise.

2. Creativity is subjective. What feels new or inspiring to you is valid, even if it's not groundbreaking in a global or historical context.

3. Human-centered AI design helps artists and creatives by taking care of repetitive tasks, freeing them to focus on more meaningful work.

Discussion Questions

1. The podcast distinguishes between P-creativity (personal) and H-creativity (historical). When you use an AI tool, like an image generator or a writing assistant, do you feel it's a creative process?

2. The podcast mentions that AI can handle "busy work" like converting files. In your field, what are some repetitive or mundane tasks that an AI could take over?

3. What are the potential legal and ethical challenges in determining the ownership of a piece of art or music created with the help of an AI? Should the user, the AI developer, or the original creators whose data was used (to teach the AI model) have a claim to the work?

Relevant Resources

For more information on topics mentioned in this podcast, check out:

- Richard Sutton's work on deep reinforcement learning
  - "The Bitter Lesson" essay on incompleteideas.net.
  - Amii.ca articles and interviews on the Amii website.
- Margaret Boden's work on computational creativity
  - "Computer Models of Creativity" on ResearchGate.

References

Amii. (2024, May 28). *Approximately correct: The deal with disinformation* [Video]. Alberta Machine Intelligence Institute. Retrieved October 1, 2025, from https://www.amii.ca/videos/approximately-correct-disinformation

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